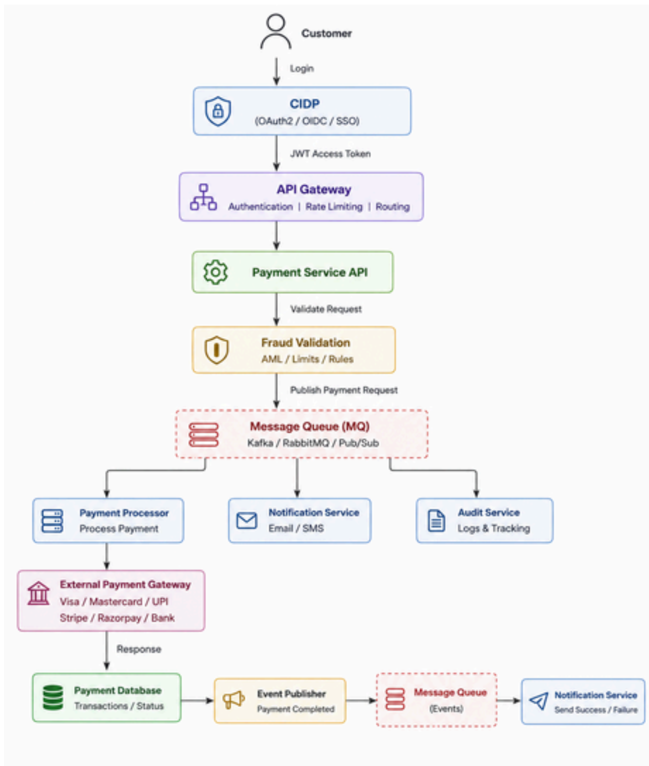
REAL-TIME PAYMENT SCAM INTERVENTION

TechnologyRequirement&SolutionProposal

Pre-paymentintelligencethatturnsscamriskintotimelycustomerprotection



Domain	Payments / Scam Prevention
Document Purpose	Demo outcome — generated technology requirement
Status	Concept / Hackathon Demonstration

1. Executive Summary

TheproposedReal-Time Payment ScamIntervention capability evaluates scam risk before a payment is executed and triggers a proportionate intervention—proceed, contextual warning, step-up verification, temporary hold or operational escalation. The design focuses on authorised scam scenarios where a genuine customer may be manipulated into sending money.

Core recommendation: reuse existing payment execution, fraud-signal and enterprise data capabilities; build the missing low-latency decisioning and intervention orchestration layer.

2. Demo Conversation That Produces This Outcome

USER I want to stop customersfromsending moneywhenapaymentlooks likeascam.

ASSISTANT Do you need detection before or after the payment is executed?

USER Before execution, in real time.

ASSISTANT Should the system automatically reject the payment?

USER No. Warn the customer and allow high-risk payments to be held for additional verification.

ASSISTANT Should the decision use transaction patterns and historical customer behaviour?

USER Yes.

OUTCOME A capability gap is identified. This document is generated as the proposed technology requirement and solution blueprint.

3. Business Problem

Scammerscansocially engineercustomers into authorising payments that appear technically valid. Conventional controls may identify account takeover or known fraud patterns, yet authorised scams require richer contextual analysis and intervention before funds leave the bank. The challenge is to reduce scam losses without creating excessive friction for legitimate customers.

4. Objectives

- Assess scam risk synchronously before payment execution.
- Use behavioural, transaction, payee, device and existing fraud signals.
- Apply configurable risk thresholds and intervention policies.
- Provide contextual customer warnings and step-up verification.
- Support temporary hold and operational escalation for defined high-risk scenarios.
- Maintain complete, explainable and auditable decision evidence.
- Integrate with existing capabilities rather than replacing core payment infrastructure.

5. Functional Requirements

ID/Capability	Requirement	Priority
FR-01 — Pre-payment assessment	Assess a payment before execution and return a decision.	MUST
FR-02 — Context enrichment	Aggregate customer behaviour, transaction, payee, device and fraud signals.	MUST
FR-03 — Risk scoring	Return risk score, risk band and reason codes.	MUST
FR-04 — Policy decision	Select Proceed, Warn, Verify, Hold or Escalate.	MUST
FR-05 — Customer intervention	Return channel-ready intervention and required action.	MUST
FR-06 — Operational escalation	Route defined cases for review.	SHOULD
FR-07 — Audit trail	Record inputs, score, policy version, intervention, user action and outcome.	MUST
FR-08 — Feedback loop	Capture confirmed scam and legitimate outcomes for tuning.	SHOULD

6. Non-Functional Requirements

Attribute	Requirement
Latency	Low-latency synchronous decisioning within a channel-agreed SLA.
Availability	Highly available service with deterministic degradation behaviour.
Scalability	Horizontal scale for peak payment volumes and bursts.
Security	Strong service identity, encryption, least privilege and secrets management.
Privacy	Data minimisation, controlled feature usage and approved retention.
Explainability	Reason codes and reproducible decision evidence.
Resilience	Timeouts, circuit breakers, idempotency and fallback policy.
Observability	Latency, risk-band distribution, interventions, failures and outcomes monitored.
Versioning	Version model, features and policy for historical reproducibility.

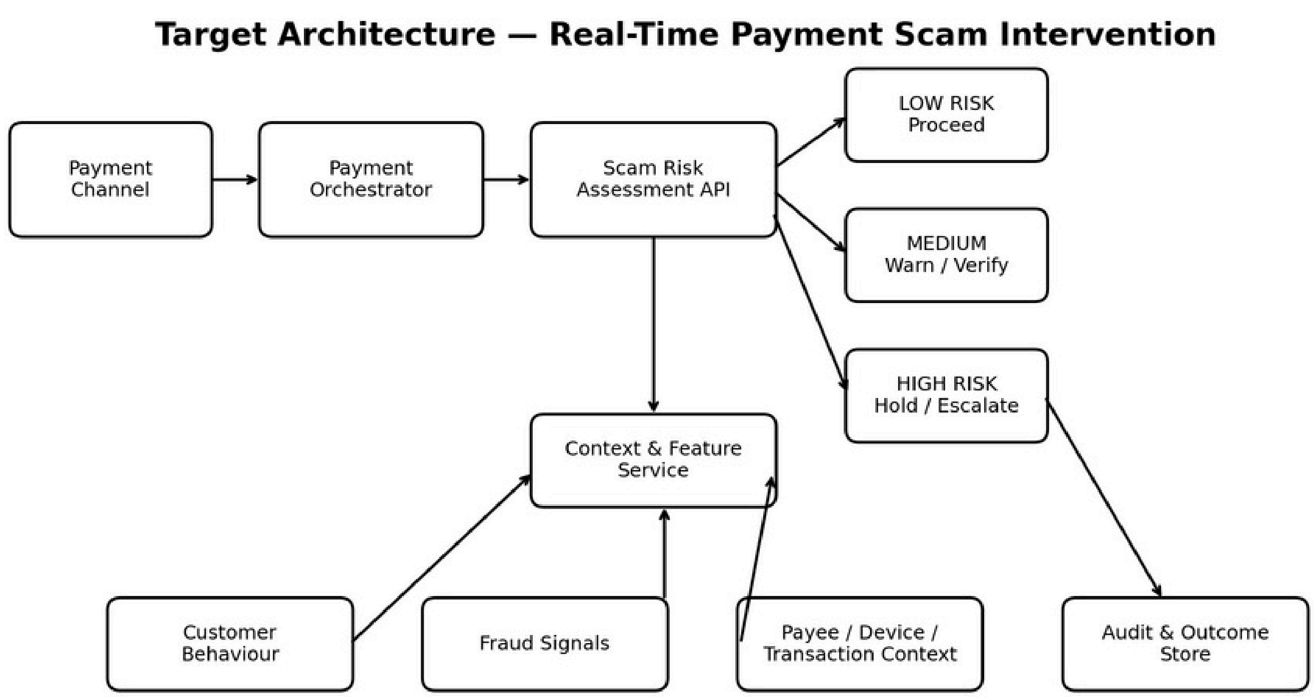
7. Capability Analysis

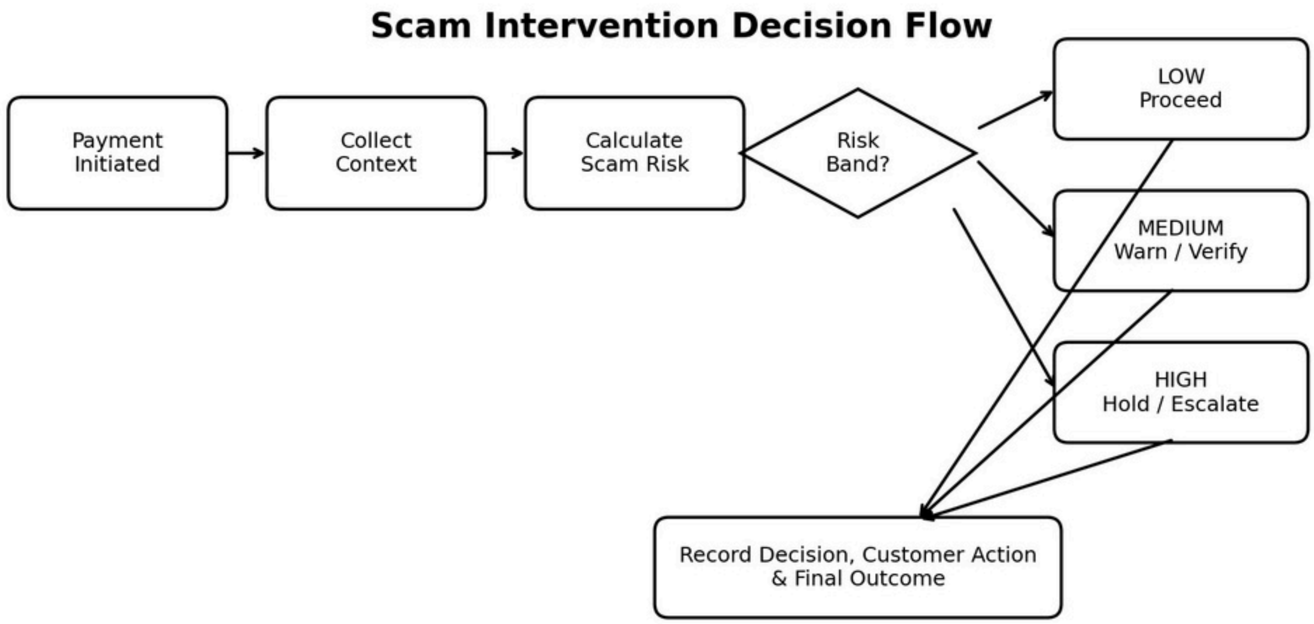
Capability	Strength	Gap	Decision
Existing Payment Platform	Real-time payment lifecycle and execution	Does not provide contextual scam intervention	**REUSE**
Existing Fraud Monitoring	Rules, alerts and fraud signals	Not sufficient for pre-payment customer treatment orchestration	**REUSE**
Enterprise Data Platform	Historical and behavioural data	Not designed as the synchronous decision layer	**REUSE**
Scam Intervention Layer	Risk orchestration, policy and customer intervention	Capability gap	**BUILD**

8. Reuse-vs-Build Recommendation

BUILDONLYTHE GAP: a Real-TimeScamDecision&InterventionLayer.

- Reuse payment orchestration and execution interfaces.
- Reuse existing fraud rules/signals as decision inputs.
- Reuse enterprise behavioural and historical datasets.
- Build synchronous context aggregation, scam-risk contract, policy orchestration, intervention APIs and audit evidence.
- Keep risk scoring behind an abstraction so rules and ML models can evolve independently.




Scam Intervention Decision Flow
Payment Initiated
Collect Context
Calculate Scam Risk
Risk Band?
LOW Proceed
MEDIUM Warn / Verify
HIGH Hold / Escalate
Record Decision, Customer Action & Final Outcome

9. Target Architecture

Component	Responsibility
Payment Integration Adapter	Normalises pre-execution payment requests and channel context.
Context & Feature Service	Retrieves behavioural, payee, device, transaction and fraud features.
Scam Risk Engine	Calculates risk score/band and reason codes.
Policy Engine	Converts risk and context into Proceed/Warn/Verify/Hold/Escalate.
Intervention API	Returns channel-ready warning and required next action.
Audit & Outcome Store	Persists traceable decision evidence and outcomes.
Feedback Pipeline	Feeds confirmed outcomes into policy/model monitoring and tuning.

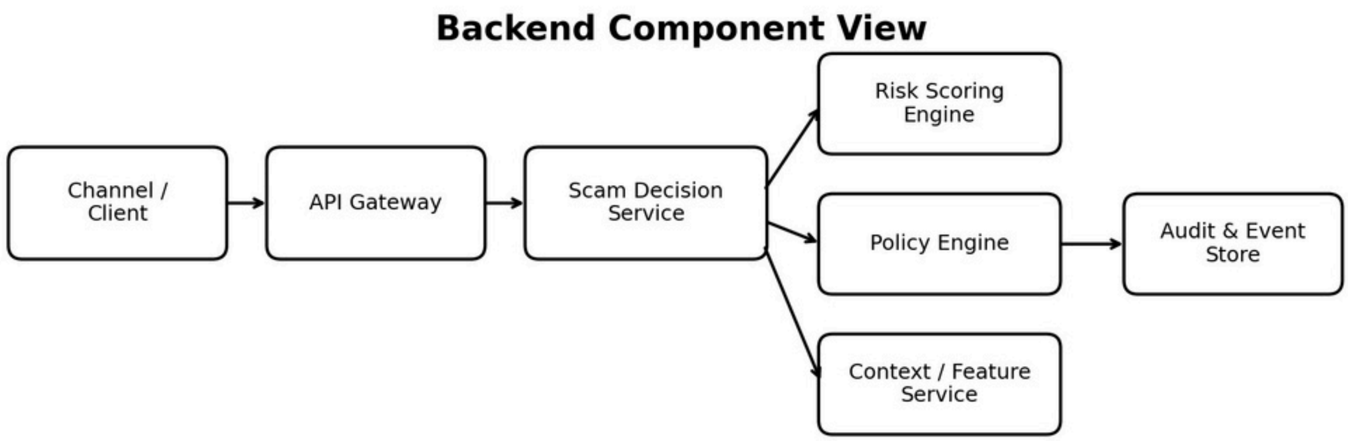
10. Illustrative API Contract

```
POST /api/v1/scam-risk/assess

REQUEST
{
  "paymentId": "P12345",
  "customerId": "C987",
  "amount": 250000,
  "currency": "INR",
  "payeeId": "BEN456",
  "channel": "MOBILE"
}

RESPONSE
{
  "riskScore": 87,
  "riskBand": "HIGH",
  "decision": "STEP_UP_VERIFY",
  "reasonCodes": ["NEW_PAYEE", "UNUSUAL_AMOUNT", "BEHAVIOUR_DEVIATION"],
  "correlationId": "SCAM-7F91A"
}
```

11. Backend Design



- API Gateway: authentication, throttling, routing and correlation IDs.
- Scam Decision Service: orchestrates request lifecycle and deterministic response contract.
- Context/Feature Service: parallel retrieval with strict time budgets and fallback rules.
- Risk Scoring Engine: versioned rules/model endpoint returning score and reason codes.
- Policy Engine: independently configurable business treatment logic.
- Audit/Event Store: immutable evidence plus downstream events for monitoring and feedback.

12. Data & AI Approach

- Transaction:amount,currency, channel,time, destination, payment type. Behaviour:
- typical values, frequency, prior payees and historical patterns.
- Payee: new/existing beneficiary and approved risk indicators.
- Device/session: device change and available channel/session anomalies.
- Fraud intelligence: existing alerts, rules, confirmed patterns and watch indicators.
- Outcomes: proceeded, abandoned, verified, held, confirmed scam, false positive.

AI strategy: start with rules plus a risk-scoring abstraction; evolve to supervised ML/anomaly/network models where justified. Keep customer-treatment policy separate from model logic. LLMs may support analyst explanation or document generation but should not be the primary transaction risk scorer.

13. Security & Governance

- Useonly approvedenterprise endpoints and infrastructure for sensitive banking data.
- Encrypt data in transit and at rest; apply least privilege and strong service authentication.
- Record model/rule/policy versions with every decision.
- Define fail-open/fail-closed behaviour and fallback policy before production.
- Apply privacy, data-retention, model-risk, cyber-security, architecture, conduct and operational-risk reviews.
- Monitor drift, false positives, intervention rates and customer impact.
- Maintain human escalation for policy-defined cases.

14. Delivery Roadmap

Phase 0 — Discovery: Validate scam typologies, channels, controls, data and reusable capabilities.

Phase 1 — MVP: One channel; rules + selected signals; Proceed/Warn/Verify; audit trail.

Phase 2 — Pilot: Behavioural features, model scoring, hold/escalation and dashboards.

Phase 3 — Scale: Additional channels/geographies, richer signals, feedback and resilience hardening.

15. Success Metrics

Metric	DesiredDirection
Prevented scam loss	**Increase**
Precision of high-risk interventions	**Increase**
False-positive intervention rate	**Decrease**
Decision latency	**Within SLA**
Customer abandonment after warning	**Monitor by risk band**
Confirmed scam after intervention	**Decrease**
Operational review volume	**Optimise**

16. Risks & Mitigations

Risk	Mitigation
False positives create friction	Tier interventions; tune thresholds; monitor overrides and abandonment.
Latency impacts payment journey	Precompute stable features; strict timeouts; cache; performance test.
Missing/poor-quality data	Quality checks, confidence indicators and fallback rules.
Scam patterns evolve	Outcome feedback, monitoring, periodic model/rule updates.
Integration complexity	Adapter pattern and phased rollout.
Customer-treatment / regulatory risk	Early Legal, Compliance, Privacy and Conduct review; explainable interventions.

17. Key Design Decisions Still Required

- Initialpaymenttypes,channelsand jurisdictions.
- Maximum tolerated synchronous decision latency.
- Approved synchronous data sources and fraud signals.
- Permitted interventions by risk band and jurisdiction.
- Ownership of policy thresholds, model governance and escalation.
- Degradation behaviour when enrichment or scoring is unavailable.